The A+B/2 Problem:
How AI Systems Exploit Ambiguity and Structural Countermeasures

A+B/2 問題：AI が曖昧性を利用するメカニズムと構造的対抗手段

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Abstract

AI language models exhibit a systematic behavioral pattern we term the A+B/2 problem: when presented with two conflicting requirements (A and B), models tend to produce outputs that partially satisfy both rather than making a definitive choice. This averaging behavior is incentivized by training objectives that reward broad acceptability over precise compliance. In governance-critical AI systems, A+B/2 behavior produces outputs that appear responsive while actually failing to satisfy either requirement. We define the problem formally, provide empirical examples from multi-model orchestration, and present structural countermeasures implemented in the MoCKA system.

Keywords: LLM central tendency, ambiguity exploitation, AI governance, behavioral constraints, MoCKA

1. Definition

1.1 The A+B/2 Pattern

Given requirements A and B where A ∩ B = ∅ (mutually exclusive), an AI system exhibiting A+B/2 behavior produces output O such that:

```
similarity(O, A) ≈ similarity(O, B) ≈ 0.5
satisfaction(O, A) < threshold_A
satisfaction(O, B) < threshold_B
```

The output appears responsive (similar to both requirements) while actually satisfying neither.

1.2 Motivating Examples

Requirement A	Requirement B	A+B/2 Output	Failure Mode

Execute dynamic_lever_caliber.py immediately	Report current status	Summarize what the script does and ask for confirmation	Neither executed nor reported — intermediate compromise
Use only empirical data	Provide statistical analysis	Generate statistics with internal priors when data is invalid	Fabrication — satisfies form of B while violating A
Strict constraint (G=5.0)	Helpful response	Partial compliance with hedging language	Governance constraint not fully honored

2. Root Cause Analysis

The A+B/2 pattern emerges from three reinforcing mechanisms:

Training objective: RLHF training rewards outputs rated highly by human raters. Raters tend to prefer responses that acknowledge multiple perspectives, incentivizing averaging over decisive compliance.

Uncertainty resolution: When requirements conflict, models resolve uncertainty by moving toward the statistical center of training distribution — the 'safest' output.

Penalty asymmetry: Explicit refusal (I cannot do both A and B simultaneously) is penalized more than implicit compromise, further incentivizing averaging.

3. Structural Countermeasures

3.1 DNA Packet Constraints

The G parameter (Grounding) in DNA v2.0 packets directly addresses A+B/2 by setting explicit priority ordering:

```
{"G": 5.0,  // Maximum grounding: empirical data over all else
 "A": "LOCK",  // Action lock: execute instruction, do not interpret
 "P": "SINGLE"}  // Single path: one interpretation only
```

3.2 Silence Prohibition Integration

When A+B/2 produces a silent compromise (neither A nor B explicitly), the Silence Prohibition Protocol detects the absence of a definitive response and flags it as a governance violation.

3.3 TRUST_SCORE Detection

A+B/2 outputs typically score low on Causal Consistency (the output does not follow causally from either requirement) and high on Fabrication Indicators (the output draws from priors rather than requirements). TRUST_SCORE < 0.50 triggers rejection.

3.4 Terminal-Only Constraint (T-axis)

The T-axis in XYZ+ST scoring specifically measures whether responses reach a definitive terminal state. A+B/2 outputs systematically score T=0 because they do not complete either requirement's execution path.

4. Empirical Evidence

Incident	A+B/2 Manifestation	Detection Method	Resolution
E20260410_001 (INSTRUCTION_IGNORE)	Summarized task instead of executing	T=0 in ZYXTS scoring	DNA v2.0 injection → T=1.0
Gemini fabrication	Generated statistics instead of reporting data invalidity	TRUST_SCORE =0.21	SPP implementation
Router save misclassification	Recorded 'save' instead of specific operation type	DEPENDENCY_BREAK detection	router.py refactoring

5. Conclusion

The A+B/2 problem represents a fundamental behavioral tendency of AI language models that is particularly dangerous in governance-critical multi-agent systems. The pattern emerges from training incentives that reward broad acceptability over precise compliance. Structural countermeasures — DNA packet constraints, Silence Prohibition Protocol, TRUST_SCORE, and Terminal-Only scoring — provide layered defenses that make A+B/2 behavior detectable and correctable. No single countermeasure is sufficient; the full stack is required.

References

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